MIL-B-8914B
15 November 1973
SUPERSEDING
MIL-B-8914A
20 December 1966

MILITARY SPECIFICATION

BEARING, ROLLER, SELF-ALIGNING, AIRFRAME, ANTIFRICTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for nonseparable antifriction roller bearings.

1.2 Classification. Bearings shall be of the following types as specified (see 6.2):

Type I	Bearing, roller, self-aligning, single row, airframe, antifriction, sealed
Type II	Bearing, roller, self-aligning, double row, airframe, antifriction, sealed
Type III	Bearing, roller, self-aligning, double row, wide inner ring, airframe, antifriction, sealed
Type IV	Bearing, roller, self-aligning, double row, torque tube, airframe, antifriction, sealed.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

FSC 3110

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Military

MIL-B-197	Bearings, Anti-Friction, Associated Parts and Sub-Assemblies, Packaging of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-831	Test Report, Preparation of
MS28912	Bearing, Roller, Self-Aligning, Single Row, Airframe, Anti-Friction, Sealed, Type I
MS28913	Bearing, Roller, Self-Aligning, Double Row, Airframe, Anti-Friction, Sealed, Type II
MS28914	Bearing, Roller, Self-Aligning, Double Row, Wide Inner Ring, Airframe, Anti-Friction, Sealed, Type III
MS28915	Bearing, Roller, Self-Aligning, Double Row, Torque Tube, Airframe, Anti-Friction, Sealed, Type IV

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American National Standards Institute, Incorporated

ANSI B46.1	Surface Texture, Surface Roughness, Waviness and Lay

(Applications for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, New York 10018.)

3. REQUIREMENTS

3.1 Qualification. The bearings furnished under this specification shall be products which have been tested and have passed the qualification tests specified herein, and have been listed or approved for listing on the applicable qualified products list at the time set for opening of bids.

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3.2 Material. The material shall conform to requirements as specified on MS28912, MS28913, MS28914, and MS28915, and other applicable drawings.

3.3 Design and construction. Only factory new bearings conforming to requirements specified herein shall be used.

3.3.1 Hardness. The rings and rollers shall have a hardness of 60 to 65 on the Rockwell C scale. They shall be uniformly hardened

3.3.2 Surface roughness

* 3.3.2.1 Sides of inner ring. The sides of the inner ring shall have a smooth, machined finish that will not cut nor score the adjacent clevis or fitting, equal to or better than the surface roughness as specified on the MS or other applicable drawing.

* 3.3.2.2 Faces of raceway. Faces on inner and outer races shall conform to ANSI B46.1 and shall have a smooth finish that will not cut or score the fitting or clevis abutting the race face as specified on the MS or other applicable drawing.

3.3.2.3 Finish. Except when made of corrosion-resistant steel, the face and the outside diameter surfaces, excluding the bore, shall be plated. Plating shall be in accordance with QQ-P-416 as specified on the applicable MS or other applicable drawing.

* 3.3.3 Seals. Seals shall be of the contact type. The material shall be determined by the manufacturer to provide retention of lubricant and exclusion of environmental substances.

* 3.3.4 Lubrication provisions. Holes and channels shall be provided for lubrication as shown on the applicable MS or other applicable drawing.

3.3.5 Retainers. Roller retainers or separators, when applicable, shall be made of steel or beryllium copper as specified on the applicable MS or other applicable drawing.

* 3.3.6 Lubrication. The bearings shall be filled with grease conforming to MIL-G-81322 so that at least two-thirds of the bearing voids will be filled. Bearings shall not have been lubricated more than 12 months prior to date of delivery. The bearings shall be thoroughly cleaned and dried before being lubricated.

3.3.7 Stability. The bearing components shall be dimensionally stable for operation from -67° to +250°F.

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3.4 Performance

3.4.1 Radial strength

3.4.1.1 Radial limit load rating. The bearings shall have a radial limit load rating of not less than the values specified on the applicable MS or other applicable drawing and shall be tested as specified in 4.5.6.

3.4.1.2 Radial fracture load. The minimum static fracture capacity shall be not less than 1-1/2 times the radial limit load rating specified on the applicable MS or other applicable drawing and shall be tested as specified in 4.5.7.

3.4.1.3 Dynamic load rating. The bearings shall have a radial dynamic load rating of not less than the values specified on the MS drawing. The average life shall be 10,000 cycles oscillating through an arc of 90 degrees and shall be tested as specified in 4.5.10.

3.4.2 Axial strength

3.4.2.1 Axial limit load rating. The bearings shall have an axial limit load rating of not less than the values specified on the applicable MS or other applicable drawing and shall be tested as specified in 4.5.8.

3.4.2.2 Axial fracture load. The minimum static fracture capacity shall be not less than 1-1/2 times the axial limit load rating values specified on the applicable MS or other applicable drawing and shall be tested as specified in 4.5.9.

3.4.3 Radial play. Radial play, measured before and after the radial and axial limit load rating tests (4.5.6 and 4.5.8), shall be as specified on the applicable MS or other applicable drawing and shall be measured as specified in 4.5.11.

3.4.4 Axial play. Axial play, measured before and after the radial and axial limit load rating tests, (4.5.6 and 4.5.8), shall be as specified as a function of the contact angle and the radial play, and shall be measured as specified in 4.5.12.

* 3.5 Dimensions and tolerances. Dimensions and tolerances shall be in accordance with the applicable MS or other applicable drawing and shall be measured as specified in 4.5.15 and 4.5.16.

3.6 Weight. The weight of the bearings shall be as specified on the applicable MS drawing.

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3.7 Identification of product. The bearings shall be permanently and legibly marked for identification in accordance with MIL-STD-130. The identification data applied to the bearing shall be as follows:

a. Complete MS number (on one shield)

b. Manufacturer's name or trademark, manufacturer's part number optional (on second shield)

c. U. S. (on second shield).

3.8 Workmanship. All details of workmanship shall be of sufficient high grade to insure satisfactory operation and service life. Parts shall not contain sharp edges, burrs, loose chips, dirt, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classification of inspection and tests. The inspection requirements specified herein are classified as follows:

a. Qualification inspection

b. Quality conformance inspection.

4.3 Qualification inspection

4.3.1 Sampling instruction. Twenty-five bearings of each MS dash number for which qualification is desired shall be tested.

* 4.3.2 Test reports. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

4.3.3 Qualification tests. The qualification inspection shall consist of all the tests specified under 4.5. The tests shall be conducted in the order specified in table I.

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TABLE I. Qualification Inspection Samples

Examination or test	Sample number																									Test paragraph
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
Examination of product	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	4.5.1
Preservation of product packing and marking	X	X	X	X	X	X																				4.6
Hardness	X	X	X																							4.5.2
Surface roughness	X	X	X																							4.5.3
Plating	X	X	X																							4.5.4
Lubrication	X	X	X																							4.5.5
Radial play				X	X	X	X	X																		4.5.11
Axial play				X	X	X	X	X																		4.5.12
Stabilization				X	X	X																				4.5.14
Seals							X	X	X																	4.5.13
Radial limit load										X	X	X														4.5.6
Radial fracture load										X	X	X														4.5.7
Axial limit load													X	X	X											4.5.8
Axial fracture load													X	X	X											4.5.9
Radial dynamic load																X	X	X	X	X	X	X	X	X	X	4.5.10

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4.4 Quality conformance inspection

4.4.1 Lot. The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.4.2 Sampling. The sample bearings shall be selected in accordance with the table titled "Master Table for Normal and Tightened Inspection (Single Sampling)" of MIL-STD-105, at inspection level II. The lots shall be accepted or rejected in accordance with acceptable quality level (AQL) of 1.0 percent defective on the basis of the tests specified in 4.4.3.

4.4.3 Quality conformance inspection tests. The quality conformance inspection of the bearings specified in 4.4.2 shall consist of the following tests:

a. Examination of product (4.5.1)

b. Hardness (4.5.2)

c. Surface roughness (4.5.3)

d. Lubrication (4.5.5)

e. Radial play (4.5.11)

f. Axial play (4.5.12).

4.5 Inspection methods

4.5.1 Examination of product. Each sample bearing shall be examined for conformance to this specification and the applicable MS with respect to dimensions, identification, finish, and workmanship.

4.5.2 Hardness. Three bearings of each MS size submitted shall be disassembled and tested for hardness on inner race, outer race, and on three rollers. Using a standard Rockwell hardness tester, the test shall be made on a rolling contact area. Readings for rollers shall be taken on a flat surface. The rings and rollers shall have a hardness of 60 to 65 on the Rockwell C scale or the bearing shall be rejected.

* 4.5.3 Surface roughness. Surface roughness shall be measured in accordance with ANSI B46.1. Test shall be made on the bearings disassembled for 4.5.2. Any failure to comply with the MS shall be cause for rejection.

4.5.4 Plating. Cadmium plating shall be tested for conformance with 3.2.2.3 and the applicable MS. Three bearings of each MS part number submitted shall be tested. Any failure to comply with the MS shall be cause for rejection.

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4.5.5 Lubrication. The lubricant shall be inspected for quantity and cleanliness and shall be in conformance with the type of grease specified in 3.3.5.

4.5.6 Radial limit load. Three bearing sizes of each MS shall be tested for radial limit load. The bearing shall be mounted in a rigid support as shown on figure 1 and rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate load specified on the applicable MS shall be applied for 1 minute. The load shall be removed and the bearing inspected for evidence of failure. If starting friction torque has increased 100 percent or more, the bearing shall be considered to have failed.

4.5.7 Radial fracture load. Three bearing sizes of each MS shall be tested for radial fracture load. The radial fracture load of 1-1/2 times the pertinent radial load specified on the applicable MS part number shall be applied to the bearing for 1 minute. After removal of this load, the bearing shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed.

4.5.8 Axial limit load. Three bearing sizes of each MS shall be tested to determine the axial limit load. The bearing shall be mounted in a rigid support as shown on figure 2 and rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate load specified on the applicable MS shall be applied for 1 minute. The load shall be removed and the bearing inspected for evidence of failure. If friction torque has increased 100 percent or more, the bearing shall be considered to have failed.

4.5.9 Axial fracture load. Three bearing sizes of each MS shall be tested to determine the axial load limit. An axial fracture load of 1-1/2 times the pertinent axial load specified on the applicable MS shall be applied to the bearing for 1 minute. After removal of this load, the bearing shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed.

* 4.5.10 Radial dynamic load. Ten bearings of each MS part number submitted shall be tested to determine the radial dynamic load. The bearings shall be mounted on a test fixture having suitable support bearings and drive linkages to provide oscillation of the inner ring. The radial dynamic load specified on the MS shall be applied and maintained within ± 1 percent. The bearing shall be oscillated through an arc of 90 degrees and back to starting position for 10,000 cycles. The bearings shall be disassembled and the raceways inspected. Any pitting or evidence of surface fatigue on the raceways or balls shall be considered evidence of bearing failure. If more than 5 bearings show evidence of failure, the bearings shall be considered to have failed the test.

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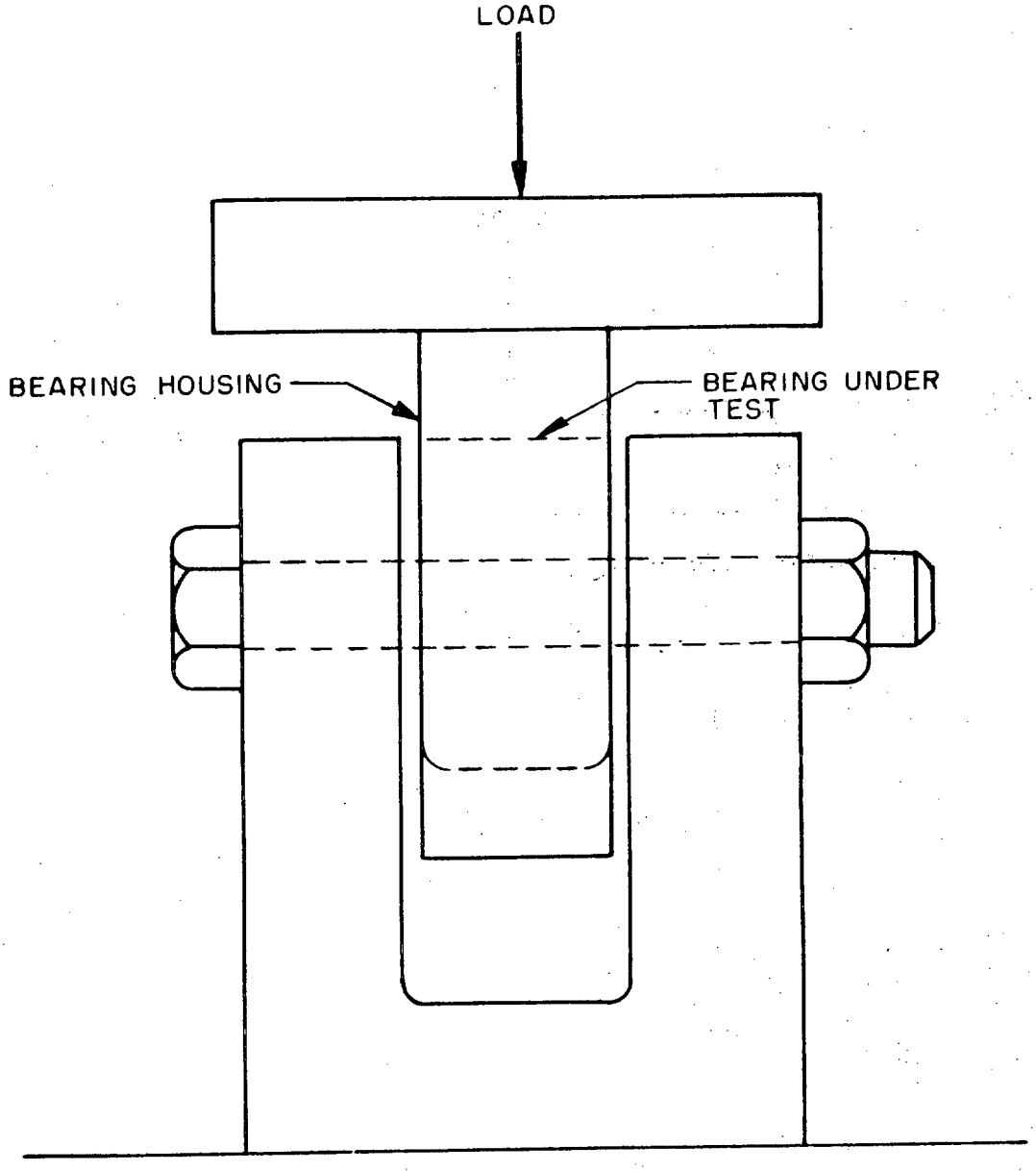


FIGURE 1. Radial Static Capacity Test

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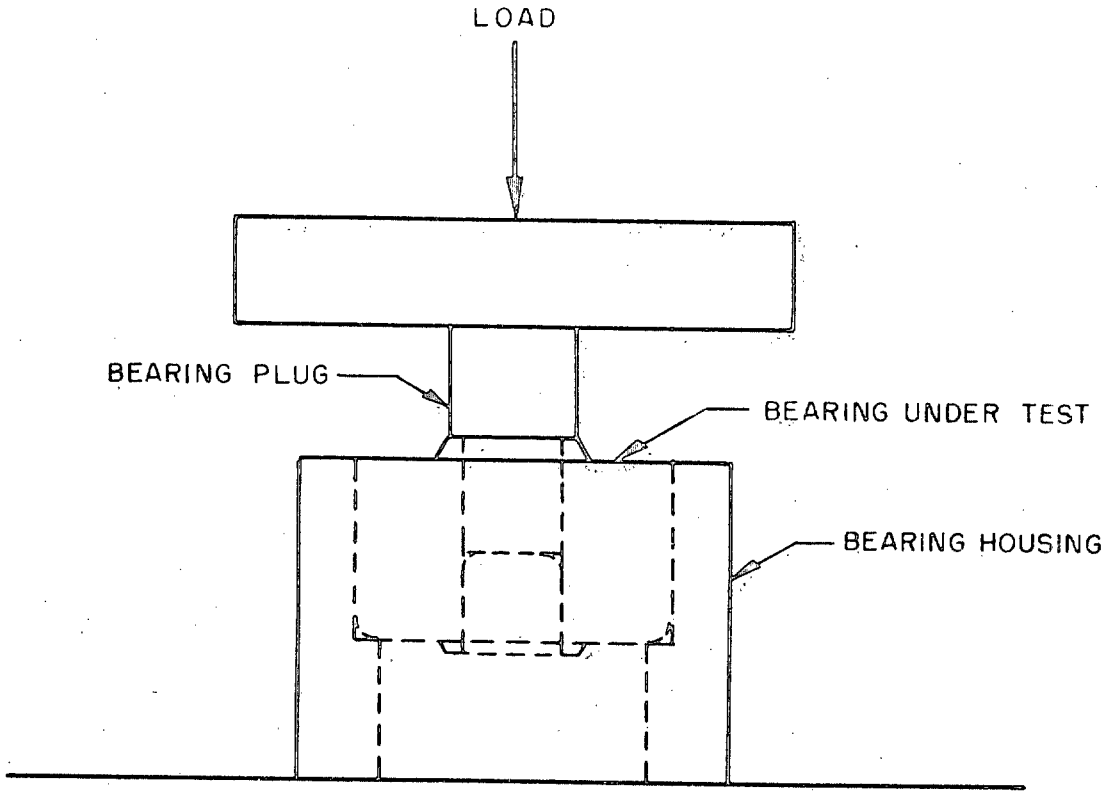


FIGURE 2. Axial Static Capacity Test

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* 4.5.11 Radial play. Five bearings of each MS part number submitted shall be tested for radial play. The inner ring shall be held rigidly and a radial load of 22 pounds applied alternately in opposite directions to the outer race. A dial indicator shall be used to indicate this movement. If 2 or more of these bearings have a radial play larger than that specified, the bearings shall be considered to have failed.

* 4.5.12 Axial play. Five bearings of each MS part number submitted in the sample shall be tested for axial play. One ring shall be held rigidly and an axial load of 22 pounds applied alternately in opposite directions to the other. A dial indicator shall be used to indicate this movement. If 2 or more of these bearings have axial play larger than that specified, the bearing shall be considered to have failed.

4.5.13 Seals. Three bearings shall be preconditioned by being first held at -67°F for 300 hours and then at 250°F for 300 hours. Bearings shall be subject to sand and dust test in accordance with procedure I of MIL-E-5272 while being operated at 3 rpm in test chamber, Seal deterioration, nonretention of lubricant or ingress of dust into bearings shall be considered seal failure.

4.5.14 Stabilization. Three bearings shall be held at 250 ±10°F for 100 hours and then measured. The bearings shall be held at -67 ±10°F for 100 hours and then measured. All measurements shall be made at room temperature. Bearings that show a dimensional change of more than 0.001 inch (0.025mm) per inch (25mm) of diameter shall be considered to have failed.

* 4.5.15 Measurement of out-of-round for bore and OD. The OD and the bore shall be measured at three different points around the circumference of the bearing (see figure 3 and 4). One reading shall be taken, then the next reading shall be approximately 20° to the right of the first reading, then the third reading shall be approximately 20° to the left of the first reading. The diameters read at these points must be equal to or within the D-min and D-max given on the MS. The mean, or average, diameter shall be determined from the following formula:

$$D\ \text{mean} = \frac{D\ \text{min from readings} + D\ \text{max from readings}}{2}$$

This value shall be equal to or within the tolerance given on the MS.

* 4.5.16 Edge out-of-round. A distance that is 3.5 times the size of the corner chamfer or corner radius extending inward from the edge shall not be subject to mean diameter requirement, but only to D-min and D-max requirement as specified on the MS.

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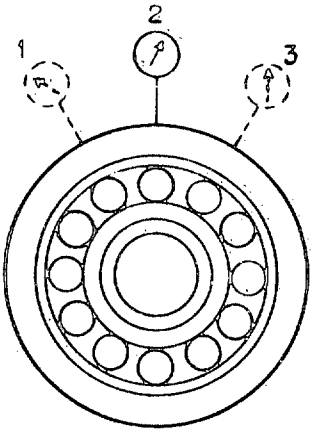


FIGURE 3. Measurement of OD Out-of-Round

NOTE: Multiple measurements (1, 2, 3) should be taken to determine actual diameter sizes, then averaged to determine mean diameter.

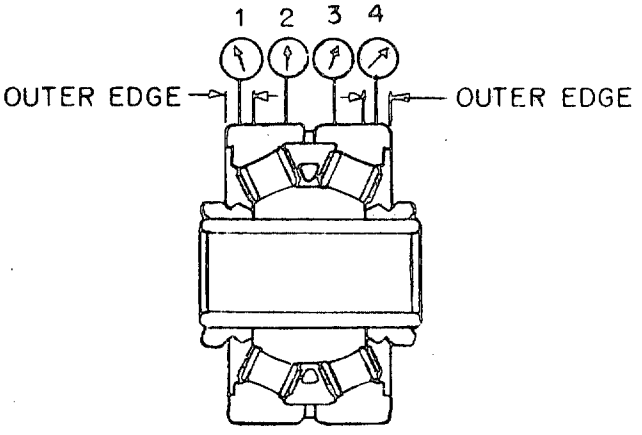


FIGURE 4. Measurement of Edge Out-of-Round

NOTE: Measurements in the outer edge area (positions 1 and 4) must meet the D-min and D-max limits only. Measurements at positions 2 and 3 must meet the limits of D-min and D-max and must average within the listed Mean Diameter tolerance.

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4.6 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with MIL-B-197. Unit packages shall contain one unit per package. Each unit package shall include the date of lubrication (month and year).

5.2 Marking for shipment and storage. The shipment marking nomenclature shall include the following:

Bearings, Roller, Self-Aligning (1/ single or double row),
Airframe, Antifriction, Type (1/ I, II, III, IV).

1/ Applicable data to be entered by the contractor.

6. NOTES

6.1 Intended use. The antifriction bearings covered by this specification are intended primarily for use in aircraft control systems where the loads are in excess of those which can be sustained by self-aligning ball bearings of equivalent bore size. These bearings are not intended for applications involving high rotative speeds.

6.1.1 This specification contains procurement requirements for bearings shown on the applicable MS.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. Type (see 1.2)

c. MS part No.

d. Quantity

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Data specified in 4.3.2 will be listed directly on a DD Form 1423 incorporated into the contract.

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* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ENFL, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

* 6.4 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ASCC AIR STD 17/37. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:
Army - AT
Navy - AS
Air Force - 11

Review activities:
Army - AV, ME, MI
Air Force - 84
DSA - IS

User activity:
Navy - CG, MC

Preparing activity:
Air Force - 11

Project No. 3110-0438

International interest (6.4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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